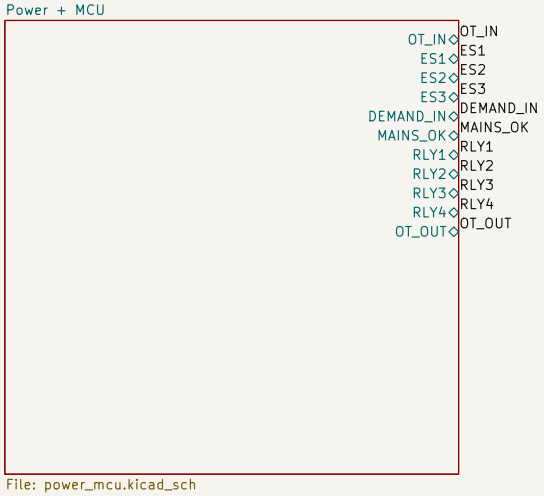
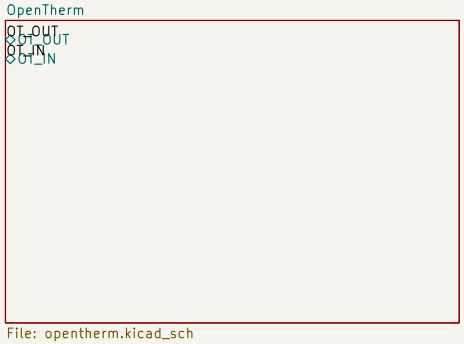
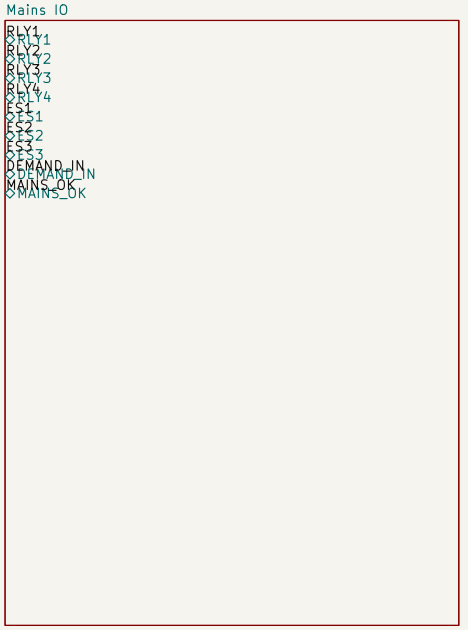




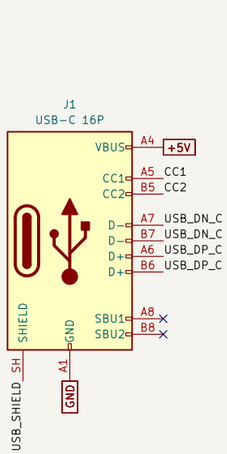


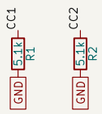
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File: heating-controller.kicad_sch

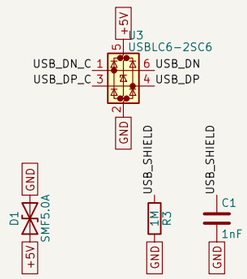
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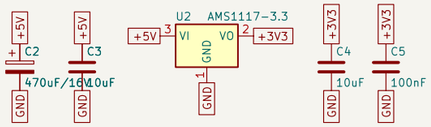
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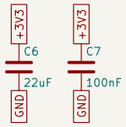
POWER + MCU. USB-C is the only supply and programmer (native USB-Serial-JTAG on IO12/IO13). Relay drives on IO21/22/23/3 only. Strapping: IO8 high (LED pull-up), IO9 BOOT, IO15 LED pull-up.

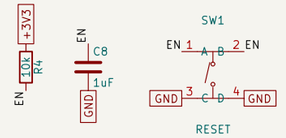


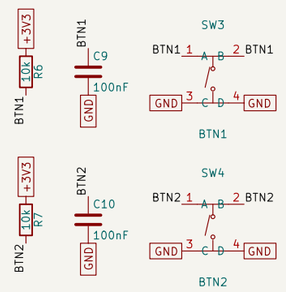


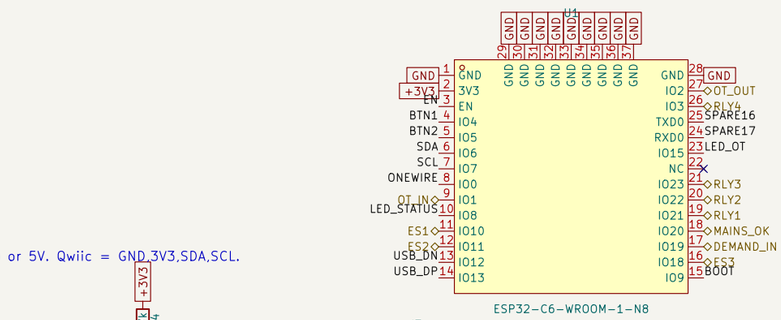




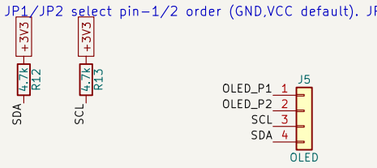




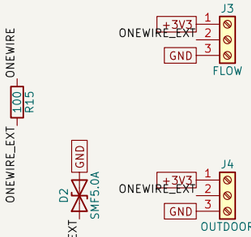


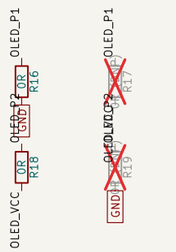


OLED header: JP1/JP2 select pin-1/2 order (GND,VCC default). JP3 selects OLED VCC 3V3 (default) or 5V. Qwiic = GND,3V3,SDA,SCL.

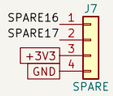


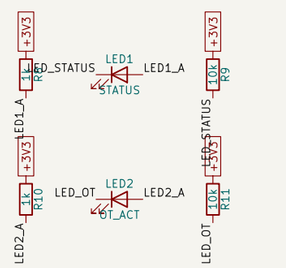








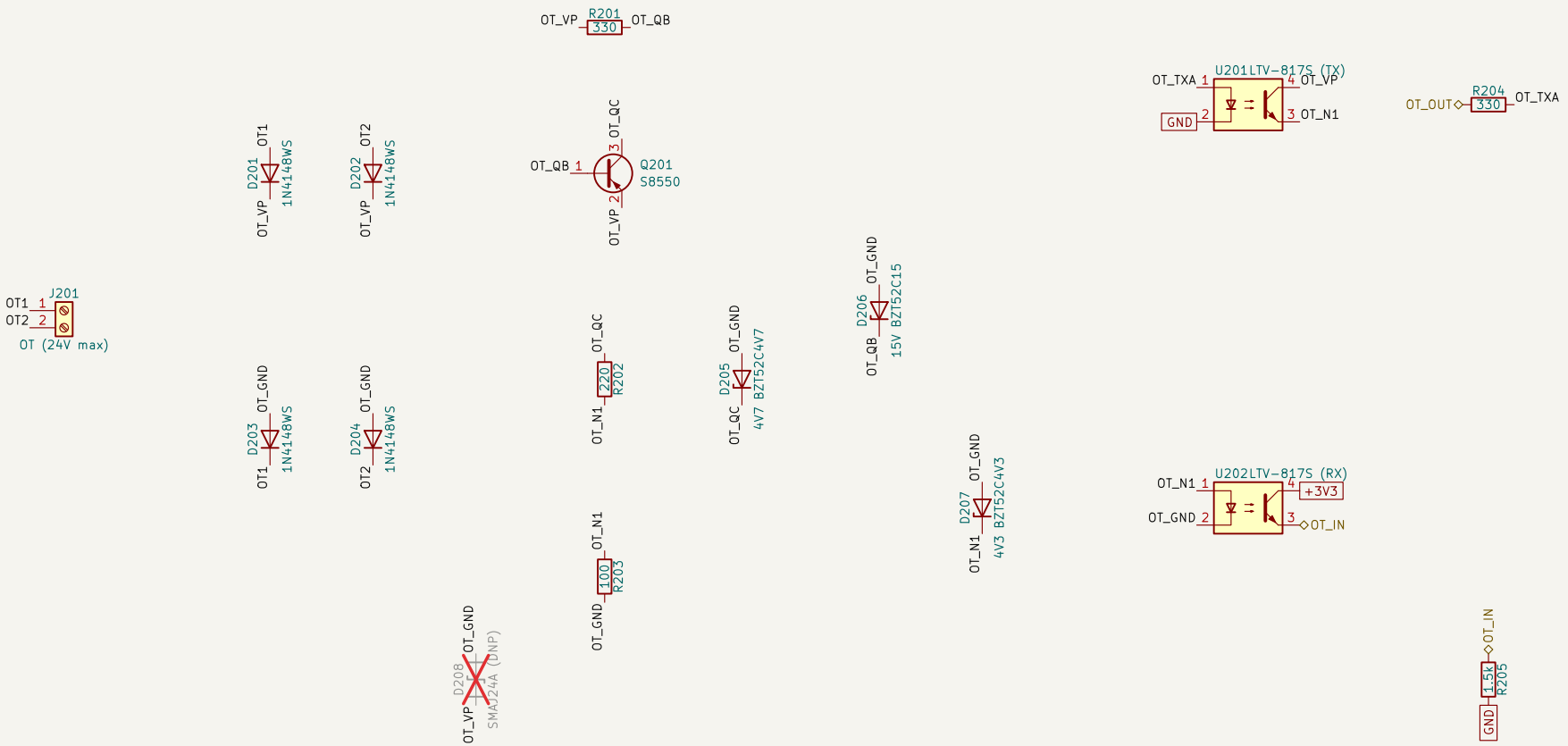






Sheet: /Power + MCU/		
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OPENTHERM MASTER ADAPTER (Ihor Melnyk topology). Bus is boiler-powered SELV, polarity independent. OT_OUT high -> U201 on -> extra bus current (TX). Boiler voltage high -> D206 conducts

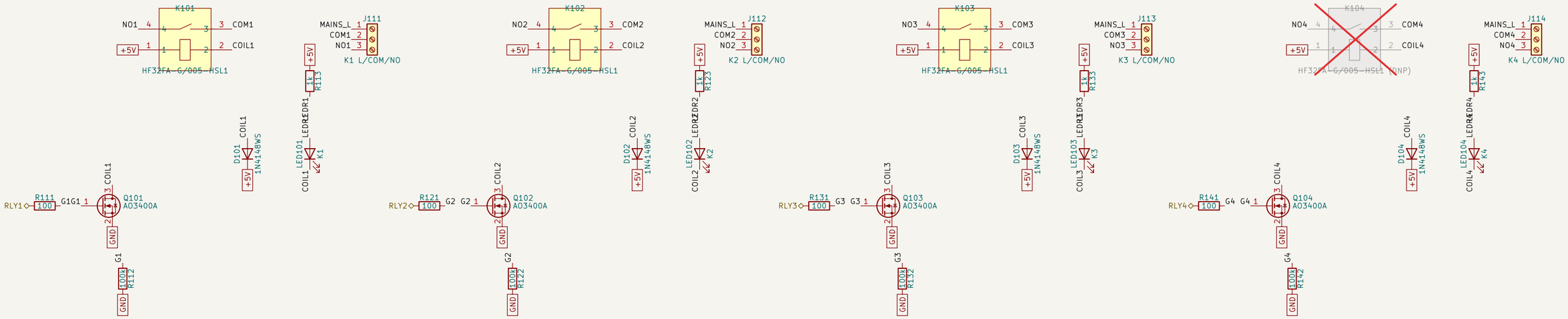


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File: opentherm.kicad_sch

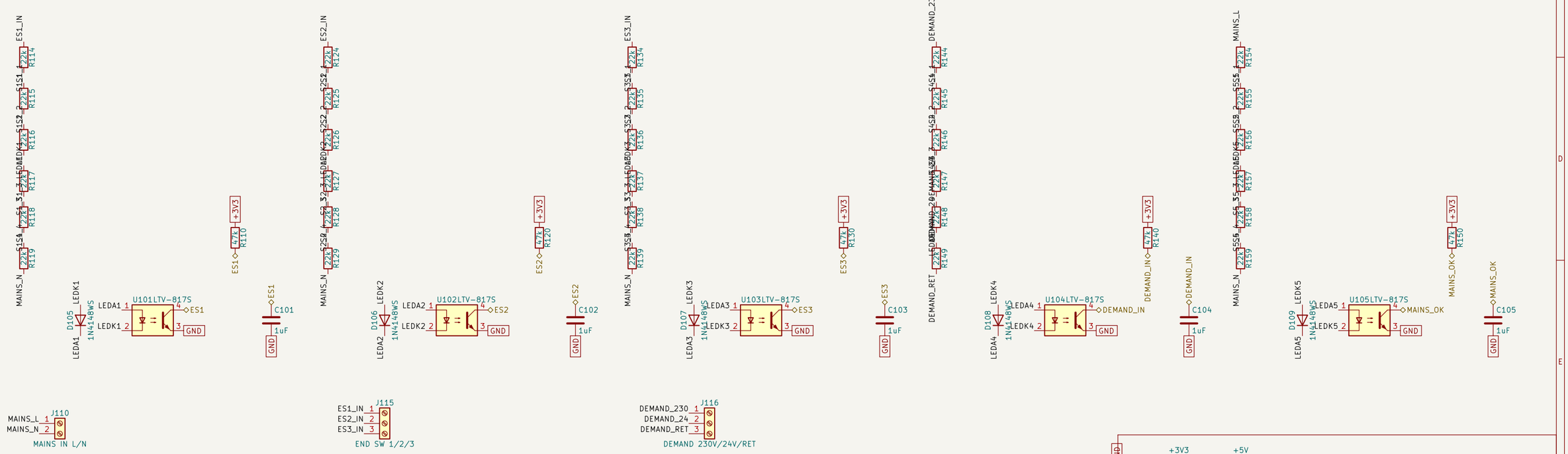
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MAINS IO. Relays K101–K104 switch MAINS_L to NO; link L–COM with a wire in the terminal for switched–live, or leave open for volt–free. K104 DNP. Gate pull–downs keep every relay OFF through reset/brownout.



230V SENSE INPUTS: 6 x 22k 1206 chain (~1.7mA, ~420V impulse per part) -> LTV-817S LED with antiparallel diode. Output is ACTIVE-LOW when live present (inverted: true in ESPHome). ES1–3 return via MAINS_N. DEMAND: use 230V or 24V input with its own RET. MAINS_OK senses board L.



Sheet: /Mains IO/
File: mains_io.kicad_sch

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